Extensions

Let $U \subset \mathbb{R}^n$ be a bdd open set st $\partial U \in C^1$ and let $1 \leq p < \infty$.

Thm: For any open set $V \subset \mathbb{R}^n$ st $\bar{U} \subset V$, $\exists$ a bdd linear operator $E : W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$ st for all $u \in W^{1,p}(U)$,

(i) $Eu = u$ a.e. in U,

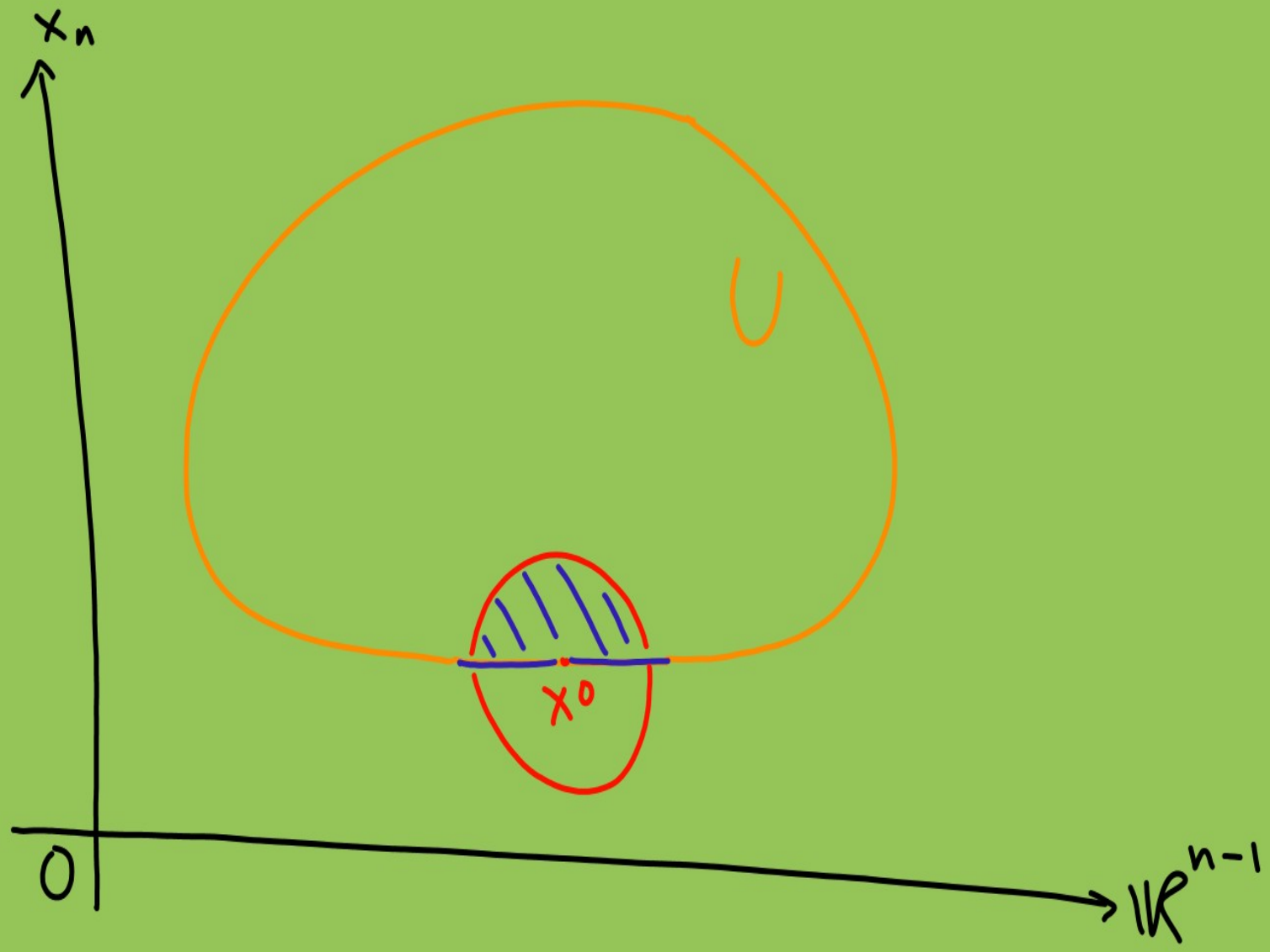
(ii) $\operatorname{spt} u \subset V$.

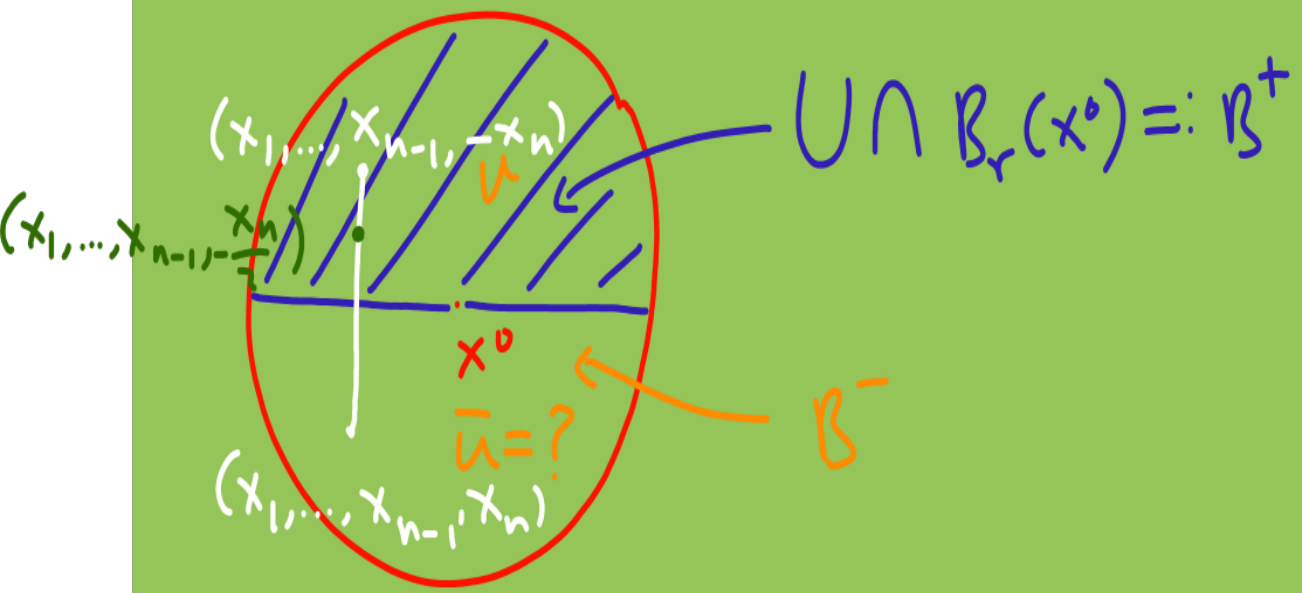
Here E is bdd means that $\exists$ a constant $C>0$ st

$$\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U).$$

<u>Pf:</u> First suppose $u \in C^1(\bar{U})$. We first consider the case where ∂U is "flat" near a point $x^0 \in \partial U$, i.e., $\exists r>0$ st

$$U \cap B_r(x^0) = \{x=(x_1,\dots,x_n) \in B_r(x^0) : x_n > 0\}.$$


x_n
U
x_0
0
$\mathbb{R}^{n-1}$



Define

$$\bar{u}(x_1,\dots,x_n) = \begin{cases} u(x_1,\dots,x_n) & \text{if } x_n \geq 0 \\ u(x_1,\dots,x_{n-1},-x_n) - 4u\left(x_1,\dots,x_{n-1},-\frac{x_n}{2}\right) & \text{if } x_n < 0. \end{cases}$$

Then

$$u_{x_n}(x_1,\dots,x_n) = \begin{cases} u_{x_n}(x_1,\dots,x_n) & \text{if } x_n \geq 0 \\ -u_{x_n}(x_1,\dots,x_{n-1},-x_n) + 2u_{x_n}\left(x_1,\dots,x_{n-1},-\frac{x_n}{2}\right) & \text{if } x_n < 0. \end{cases}$$

so $\bar{u} \in C^1(B_r(x^0))$ and

$$\|\bar{u}\|^p_{W^{1,p}(B_r(x^0))} = \|\bar{u}\|^p_{L^p(B_r(x^0))} + \sum_{i=1}^{n} \|\bar{u}_{x_i}\|^p_{L^p(B_r(x^0))}$$

$$\leq C\|u\|^p_{W^{1,p}(U)}$$

for some constant $C>0$, independent of u.

$$\|\bar{u}\|^p_{L^p(B_r(x^0))} = \int_{B_r(x^0)} |\bar{u}|^p \, dx$$

$$= \underbrace{\int_{B^+} |u|^p \, dx}_{\leq \int |u|^p dx} + \underbrace{\int_{B^-} \left|u(x_1,\dots,x_{n-1},-x_n) - 4u\left(x_1,\dots,x_{n-1},-\frac{x_n}{2}\right)\right|^p dx}_{\leq \left[\|u(x_1,\dots,x_{n-1},-x_n)\|_{L^p(B^-)}\right.}$$

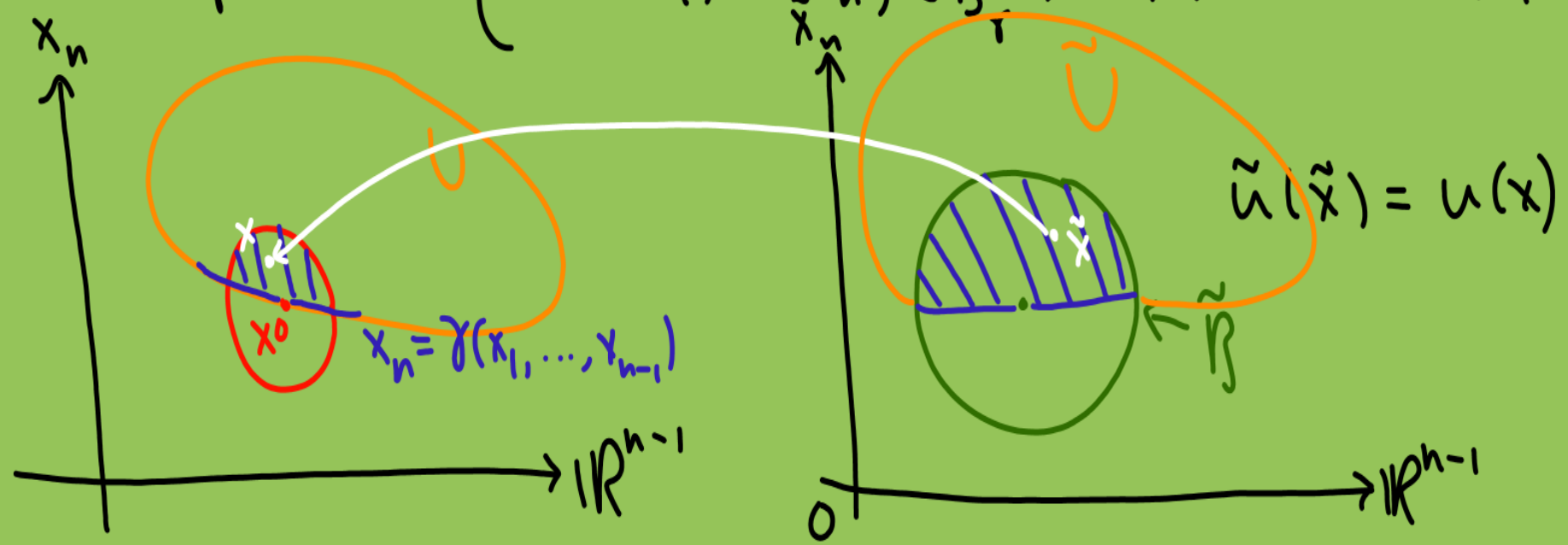
$$\leq (1+5^p)\int |u|^p dx + \left.4\left\|u\left(x_1,\dots,x_{n-1},-\frac{x_n}{2}\right)\right\|_{L^p(B^-)}\right]^p$$

Now consider the case where ∂U is not necessarily flat near x^0.

<u>Recall:</u> $\partial U \in C^1$ means that for each $x^0 \in \partial U$,

$\exists\, r > 0$ and a C^1-function $\gamma : \mathbb{R}^{n-1} \longrightarrow \mathbb{R}$ st

$$U \cap B_r(x^0) = \left\{ x = (x_1, \dots, x_n) \in B_r(x^0) : x_n > \gamma(x_1, \dots, x_{n-1}) \right\}.$$



Let $\tilde{x}_1 = x_1, \dots, \tilde{x}_{n-1} = x_{n-1}$, $\tilde{x}_n = x_n - \gamma(x_1, \dots, x_{n-1})$. Then

$$U \cap B_r(x^0) = \{\tilde{x} = (\tilde{x}_1, \dots, \tilde{x}_{n-1}, \tilde{x}_n) \in B_r(x^0) : \tilde{x}_n > 0\}.$$

Let $\tilde{u}(\tilde{x}) = u(x)$. Then $\tilde{u}$ can be extended to a function $\bar{\tilde{u}}$ st

$$\|\bar{\tilde{u}}\|_{W^{1,p}(\tilde{B})} \leq C \|\tilde{u}\|_{W^{1,p}(\tilde{U})}. \quad \text{——} \quad (1)$$

$$\tilde{u} = u(x_1, \dots, x_{n-1}, x_n - \gamma(x_1, \dots, x_{n-1}))$$

$$\tilde{u}_{x_i} = u_{x_i} - u_{x_n} \underbrace{\gamma_{x_i}}_{\text{bdd}}, \quad i = 1, \dots, n-1, \quad \tilde{u}_{x_n} = u_{x_n}$$

$|D\tilde{u}| \leq C \|Du\|$ for some constant $C > 0$ independent of u. —— (2)

Combining (1) and (2) gives

$$\|\tilde{u}\|_{W^{1,p}(\tilde{B})} \leq C \|u\|_{W^{1,p}(U)}$$

<u>Recall:</u> A linear operator $T : X \to Y$ between normed linear spaces is bdd if $\exists$ a constant $C > 0$ st

$$\|Tx\| \leq C\|x\| \quad \forall x \in X.$$

This is equivalent to the continuity of T.